

```
In [1]: def greet():
        print('hello')
        print('good morning')
```

```
In [2]: def greet():
        print('hello')
        print('good morning')
        greet()
```

hello
good morning

```
In [3]: def greet():
        print('hello team')
        greet()
```

hello team

```
In [4]: def greet():
        print('hello team')
        greet()

        def greet():
            print('hello team')
            greet()
```

hello team
hello team

```
In [5]: def greet():
        print("good evining & good night")
        greet()
        greet()
```

good evining & good night
good evining & good night

```
In [6]: def add(x,y):
        c=x+y
        print(c)
        add(5,6)
```

11

```
In [7]: def add(x, y):
        c=x+y
        print(c)
        add(5)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[7], line 4
      2     c=x+y
      3     print(c)
----> 4     add(5)

TypeError: add() missing 1 required positional argument: 'y'
```

```
In [8]: def add(x, y):
        c=x+y
```

```
print(c)
add(5, 6, 7, 8, 9)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[8], line 4
      2     c=x+y
      3     print(c)
----> 4 add(5, 6, 7, 8, 9)

TypeError: add() takes 2 positional arguments but 5 were given
```

```
In [9]: def add(x, y):
        c=x+y
        print(c)
        add(10, 20)
```

30

```
In [10]: def add(x, y, z):
          c=x+y
          print(c)
          add(10, 20)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[10], line 4
      2     c=x+y
      3     print(c)
----> 4 add(10, 20)

TypeError: add() missing 1 required positional argument: 'z'
```

```
In [11]: def add(x, y, z):
          c=x+y+z
          print(c)
          add(10, 20, 30)
```

60

```
In [12]: def greet():
          print('hello team')
          greet()

          def add(x, y, z):
              c=x+y+z
              print(c)
              add(10, 20, 30)
```

hello team
60

```
In [13]: def greet():
          print('hello team')

          def add(x, y, z):
              c=x+y+z
              print(c)

          greet()
          add(10, 20, 30)
```

```
hello team
60
```

```
In [14]: def add(x, y):
          c=x+y
          return c
          add(10, 20)
```

```
Out[14]: 30
```

```
In [15]: def add(x, y):
          c=x+y
          print(c)

          result = add(10, 20)
          print(result)
          print(type(result))
```

```
30
None
<class 'NoneType'>
```

```
In [16]: def add(x, y):
          c=x+y
          return c

          result1 = add(10, 20)
          print(result1)
          print(type(result1))
```

```
30
<class 'int'>
```

```
In [17]: def add(x, y):
          c=x+y
          return c

          def sub(x,y):
              d = x-y
              return d

          print(add(2,3))
          print(sub(2,3))
```

```
5
-1
```

```
In [18]: def add_sub(x,y): # add & sub
          c= x+y
          d= x-y
          print(c,d)

          add_sub(2,3)
```

```
5 -1
```

```
In [19]: def add_sub(x,y): # add & sub
          c= x+y
          d= x-y
          print(c,d)
```

```
r1 = add_sub(2,3)

print(r1)
print(type(r1))
```

```
5 -1
None
<class 'NoneType'>
```

```
In [20]: def add_sub(x,y): # add & sub
          c= x+y
          d= x-y
          return c,d

          r1 = add_sub(2,3)

          print(r1)
          print(type(r1))
```

```
(5, -1)
<class 'tuple'>
```

```
In [21]: def add_sub(x,y): # add & sub
          c= x+y
          d= x-y
          return c,d

          r1,r2 = add_sub(2,3)

          print(r1)
          print(r2)

          print(type(r1))
          print(type(r2))
```

```
5
-1
<class 'int'>
<class 'int'>
```

Positional Argument

```
In [23]: def person(name,age):
          print(name)
          print(age)

          person('nit', 22)
```

```
nit
22
```

```
In [24]: def person(name,age):
          print(name)
          print(age)

          person('nit')
```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[24], line 5
      2     print(name)
      3     print(age)
----> 5 person( )

TypeError: person() missing 1 required positional argument: 'age'

```

```

In [25]: def person(name,age):
          print(name)
          print(age)

          person(22)

```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[25], line 5
      2     print(name)
      3     print(age)
----> 5 person(22)

TypeError: person() missing 1 required positional argument: 'age'

```

```

In [26]: def person(name,age):
          print(name)
          print(age)

          person(22, 'nit')

```

```

22
nit

```

```

In [27]: def person(name,age):
          print(name)
          print(age+1)

          person(22, 'nit')

```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[27], line 5
      2     print(name)
      3     print(age+1)
----> 5 person(22, )

Cell In[27], line 3, in person(name, age)
      1 def person(name,age):
      2     print(name)
----> 3     print(age+1)

TypeError: can only concatenate str (not "int") to str

```

keyword argument

```

In [28]: def person(name,age):
          print(name)

```

```
print(age+1)

person(age = 22, name = 'nit')
```

```
nit
23
```

```
In [29]: def person(name,age, mob):
          print(name)
          print(age+1)
          print(mob)

          person(23, 'nit', 345)
```

```
23
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[29], line 6
      3     print(age+1)
      4     print(mob)
----> 6 person(23, , 345)

Cell In[29], line 3, in person(name, age, mob)
      1 def person(name,age, mob):
      2     print(name)
----> 3     print(age+1)
      4     print(mob)

TypeError: can only concatenate str (not "int") to str
```

```
In [30]: def person(name,age, mob):
          print(name)
          print(age+1)
          print(mob)

          person(age = 23, name = 'nit', 345)
```

```
Cell In[30], line 6
      person(age = 23, name = 'nit', 345)
                                         ^
SyntaxError: positional argument follows keyword argument
```

```
In [31]: def person(name,age, mob):
          print(name)
          print(age+1)
          print(mob)

          person(age = 23, name = 'nit', mob = 345)
```

```
nit
24
345
```

default argument

```
In [32]: def person(name,age):
          print(name)
          print(age)
```

```
person('nit')
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[32], line 5  
      2     print(name)  
      3     print(age)  
----> 5     person()  
  
TypeError: person() missing 1 required positional argument: 'age'
```

```
In [33]: def person(name, age=18):  
          print(name)  
          print(age)  
  
          person('nit')
```

```
nit  
18
```

```
In [34]: def person(name, age=20):  
          print(name)  
          print(age)  
  
          person('nit', 25)
```

```
nit  
25
```

```
In [ ]:
```